INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

POWER SOURCE, DC, CONSTANT CURRENT, SHIELDED METAL ARC WELDING (SMAW)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers constant direct current (DC) arc welding power sources intended for single operator shielded metal arc welding (SMAW) when connected to external electrical circuits as specified herein**.**

2. CLASSIFICATION. The power source shall be of the following sizes. The size to be furnished shall be as specified (see 7.3(b)).

Size 1 - 200 amperes
Size 2 - 300 amperes
Size 3 - 400 amperes
Size 4 - 500 amperes
Size 5 - 600 amperes

3. SALIENT CHARACTERISTICS

3.1 General requirements. The power source shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The power source shall include, as a minimum, the following components:

3.2.1 Power source. The power source shall generate the specified DC output current and shall maintain constant current during welding processes. Unless otherwise specified (see 7.3(c)), the power source shall include, as a minimum, front panel + and – output twist cable connections. When specified (see 7.3(d)), the power source shall have a means for power factor correction. When specified (see 7.3(e)), the power source shall have a means for line voltage compensation.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3431

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Unless otherwise specified (see 7.3(f)), the power source shall be capable of operating in parallel with two or more similar power sources of the same manufacture when connected according to the manufacturer's instructions.

3.2.2 Controls. Unless otherwise specified (see 7.3(g)), the power source controls shall, as a minimum, include an on/off power switch and a welding current selection switch. If required, the power source shall be furnished with the following as specified (see 7.3(h)):

a. Arc force control.
b. Variable hot start control.
c. Output current polarity switch.
d. Output voltage and current meters.

3.2.3 Enclosure. The power source enclosure shall be fan cooled and manufactured to withstand the rough handling normally encountered in construction areas. Ventilation holes shall be covered with screens or be small enough to prevent entry of debris. The enclosure shall be fitted with handles, wheels, or lifting eyes as specified (see 7.3(i)).

3.2.4 Electrical system. Unless otherwise specified (see 7.3(j)), the electrical system shall conform to International Electrotechnical Commission (IEC) 60974-1, "Arc Welding Equipment - Part 1: Welding Power Sources". The power source shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.3(k)). The power source external electrical circuit connection shall be a wall receptacle plug or bare wires as specified (see 7.3(l)).

3.2.5 Optional accessories. Optional accessories shall be furnished as specified (see 7.3(m)).

3.3 Machine dimensions and weight. If required, the maximum power source dimensions (length, width, and height) and weight shall not exceed the restrictions as specified (see 7.3(n)).

3.4 Safety and health requirements. The power source shall be designed and manufactured in accordance with IEC 60974-1. The manufacturer shall ensure that the power source and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR 1910, "Occupational Safety and Health Standards". If a conflict arises between the IEC 60974-1 and OSHA standards, the OSHA standards shall apply.

3.5 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(o)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.3(p)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

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3.6 Nameplate. A nameplate shall be securely attached to the power source. Unless otherwise specified (see 7.3(q)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The power source shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The power source shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The power source shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.3(r)), the power source shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.3(s)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.3(t)). Unless otherwise specified (see 7.3(u)), the machine shall be tested in accordance with environmental conditions specified in IEC 60974-1.

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5.3.2 Operational test. The power source shall be operated in accordance with the manufacturer's standard operational test procedures. The power source shall be operated for not less than 1 hour at the maximum current capacity on a 60-percent duty cycle consisting of 6 minutes of operation for each consecutive 10-minute time cycle. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The capability of the power source to weld within its specified capacity shall be demonstrated by depositing six-inch-long weld beads. Welding shall be done with 1/8- and 1/4-inch diameter, AWS-E-6011 electrodes as specified in American Welding Society (AWS) A5.1, "Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding" (DoD adopted). There shall be no evidence of difficulty in striking or maintaining the arc.

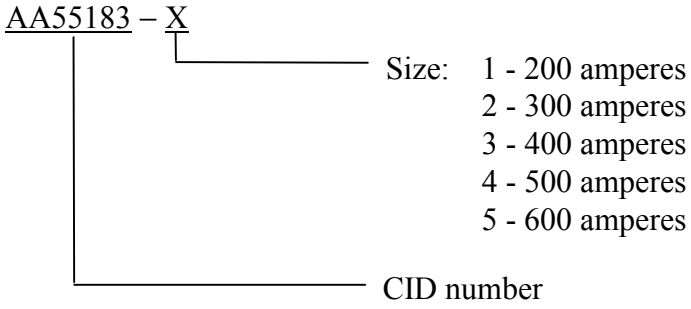
5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.3(v)).

5.4 Acceptance. Unless otherwise specified (see 7.3(w)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. Failure of the machine to meet the test requirements of section 5.3 shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.3(x)), preservation, packing, and marking shall be as specified in the order or contract.

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. A copy of the FAR may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328.

7.2.2 AWS standards. Copies of AWS standards may be obtained from the American Welding Society, 550 NW Lejeune Road, Miami, FL 33136.

7.2.3 IEC standards. Copies of IEC standards may be obtained from Global Engineering Documents, 15 Iverness Way East, Englewood, CO 80112.

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7.2.4 OSHA standards. Copies of OSHA standards may be obtained from U.S. Department of Labor, 200 Constitution Avenue NW, Room N-3649, Washington, DC 20210. Electronic copies of OSHA standards may be obtained from http://www.osha.gov.

7.3 Ordering data. Acquisition documents must specify the following:

a. CID document number, revision, and CID PIN.
b. Size of power source required (see 2).
c. Cable connections, if different (see 3.2.1).
d. Power factor correction, if required (see 3.2.1).
e. Line voltage compensation, if required (see 3.2.1).
f. Parallel operation capability, if different (see 3.2.1).
g. Controls, if different (see 3.2.2).
h. Controls, switches, and meters, as specified (see 3.2.2).
i. Enclosure handles, wheels, or lifting eyes, as specified (see 3.2.3).
j. Electrical system industrial standard, if different (see 3.2.4).
k. Electrical system voltage, as specified (see 3.2.4).
l. External electrical circuit connection, as specified (see 3.2.4).
m. Optional accessories, as specified (see 3.2.5).
n. Dimensions and weight, as specified (see 3.3).
o. System of units required (see 3.5).
p. Dual system of units, if required (see 3.5).
q. Nameplate, if different (see 3.6).
r. Testing, if different (see 5.3).
s. Test responsibility, if different (see 5.3.1).
t. Certification of calibration, as specified (5.3.1).
u. Test environmental conditions, if different (see 5.3.1).
v. Optional tests, as specified (see 5.3.4).
w. Acceptance, if different (see 5.4).
x. Packaging requirements, if different (see 6).

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99

Reviewers:
Army - CR4
Navy - AS, CG, MC, YD
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FEE

Preparing activity:
DLA - GS6

(Project 3431-0248)

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